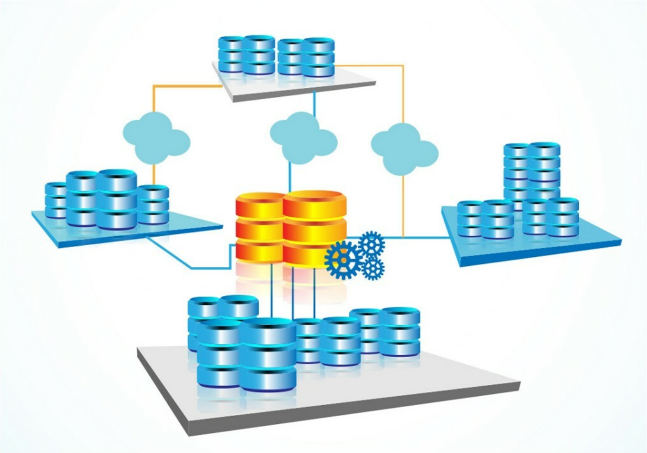
Apache Cassandra: The NoSQL Scalable Database

This article introduces readers to the Apache Cassandra NoSQL database, and provides them with use cases for which it is suitable.



Apache Cassandra is a free and open source distributed, massively scalable database management system designed to handle large amounts of data across many commodity servers, while providing highly available service and no single point of failure. Apache Cassandra offers capabilities like continuous availability, linear scale performance, operational simplicity and easy data distribution across multiple data centres and cloud availability zones.

History

Apache Cassandra was originally developed by Avinash Lakshman (one of the authors of Amazon's Dynamo) and Prashant Malik at Facebook for inbox search. Cassandra was published as an open source project in Google Code in July 2008. It was accepted into Apache Incubator in March 2009. Since February 2010, Cassandra has been an 'Apache top-level project'.

Features of Apache Cassandra

The following are the key features of Apache Cassandra.

Decentralised: There are no single points of failure and no network bottlenecks. Every node in the cluster is identical.

Supports replication and multiple data centre replication: Cassandra supports replication across multiple data centres (in multiple geographies) and multi-cloud availability zones for writes/reads.

High scalability: Read and write throughput increase linearly as new machines are added, with no downtime or interruption to applications.

Fault-tolerant: Data is automatically replicated to multiple nodes for fault-tolerance. Replication across multiple data centres is supported. Failed nodes can be replaced with no downtime.

Tunable data consistency: Cassandra supports configured consistency levels to manage availability versus data accuracy.